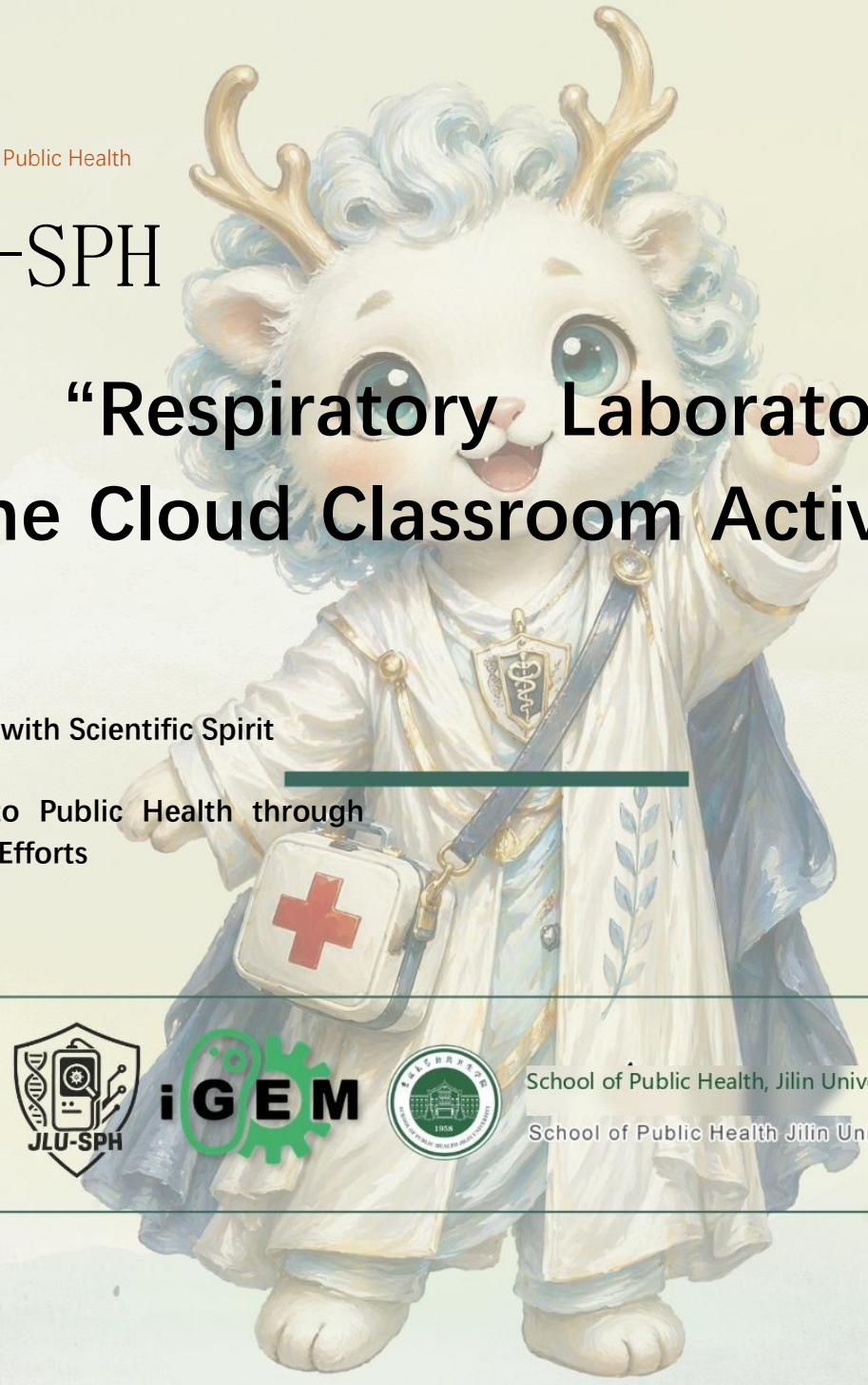
Life Protection · Public Health

JLU-SPH

2026 “Respiratory Laboratory” Online Cloud Classroom Activity Plan

Guarding Life with Scientific Spirit

Responding to Public Health through Collaborative Efforts



School of Public Health, Jilin University

School of Public Health Jilin University

JILIN UNIVERSITY · SCHOOL OF PUBLIC HEALTH

Ⅰ、 Basic Information of the Event

- Event Name: "Respiratory Laboratory" Online Cloud Class
- Event Theme: Centered on the concept of the "Microscopic Battlefield in Breathing," this course is designed for primary school students aged 6–12, featuring engaging breakdowns of synthetic biosensors and the principles of respiratory pathogen detection. The course aims to provide a lightweight, universally accessible online science education experience.
- Event Type: ☑ Educational Science Outreach ☐ Public Welfare Service ☐ Cultural Promotion ☐ Community Service ☐ Other:
- Organizer: JLU-SPH Team

Ⅱ、 Event Objectives

1. Core Objective: Leverage pre-recorded online cloud classes to circumvent constraints of time and venue, transforming complex synthetic biologyand viral detection concepts into content that is fun and understandable for young children. This addresses the shortage of frontier biological science education for primary school students, and fosters awareness of scientific epidemic prevention and technological altruism from an early age.

2. Specific Goals:

○ Knowledge/Cognition: Understand the transmission characteristics of respiratory viruses, grasp the basic logic behind micro-level virus detection using biosensors (原术语：生物传感器, biosensor), and compare the advantages of traditional detection versus rapid detection using synthetic biology.

○ Behavior/Skills: Learn to observe epidemic prevention challenges from everyday perspectives; independently propose creative, simple epidemic detection concepts; develop habits of self-directed online science learning.

○ Emotion/Attitude: Eliminate the sense of distance from microscopic organisms and laboratory technologies; appreciate the blend of scientific technique and humanistic care; proactively pay attention to vulnerable groups such as the elderly and children; foster the concept of protecting others through scientific means.

3. Expected Outcomes:

○ Number of Participants: Approximately 100 primary school students aged 6–12 will attend online.

○ Satisfaction Target: Above 90%.

○ Deliver a complete set of online science class recordings; collect post-class feedback questionnaires.

Ⅲ、Target Audience

- Core Audience: Students in grades 1–6 of primary school.
- Audience Characteristics: Deep curiosity about the microscopic world and scientific experiments; experiences with class-wide respiratory infections; difficulty understanding technical terms such as gene, biosensing ; preference for animation and story-based science education; high acceptance of online learning, not restricted by physical venues.

Ⅳ、Event Time and Location

- Event Date: June 2, 2026

- Event Duration: Approximately 20 minutes
- Event Venue:

1. Free online viewing
2. Organize a unified group viewing

Ⅴ、 Event Content and Process

[Overview of Event Process]

Stage	Time	Session Title	Specific Content	Responsible Personnel	Required Materials / Filming Focus
Stage	Three Days Before Launch	Online material production and platform debugging	Refining the presentation script, creating tiered (layered) animation PowerPoint slides, recording popular science videos, and testing the online platform playback functionality.	Main presenter, technical team	Designing PowerPoint slides, animation materials, recording equipment, and online accounts.
Main Stage	20min Online course	Layered science explanation	Use everyday analogies to deconstruct the principles of viruses, biosensors, and amplified signal detection, accompanied by animated demonstrations	Main presenter	Segmented science animations, pathogen illustrations
		Open interactive brainstorming	Guide students to brainstorm portable campus detection solutions and focus on the needs of vulnerable groups	Main presenter	Post-class creative thinking questions
Closing Stage	Three Days After Course Viewing	Questionnaire collection, creative submissions	Open online questionnaire to gather students' reflections and creative ideas	Teaching assistants	Online questionnaire link, creative submission forms

[Detailed Description of Each Phase]

- **Phase 1: Tiered Science Popularization Explanation**

Core Content: Science popularization is delivered in three progressive tiers: ① Limitations in respiratory virus transmission and detection; ② Working logic of synthetic biological sensors , with animated demonstrations illustrating the principle of amplifying trace viral signals; ③ Intelligent epidemic prevention and detection in school campuses.

  - Extension of the plan guides students to consider the convenience of use by the elderly and young children. Animation, analogies, and metaphors are employed throughout to reduce comprehension difficulty.

- **Key Message:** Even small biological detectors can capture invisible viruses, enabling us to proactively block respiratory pathogens.
- **Phase 2: Open Interactive Imagination**
  - **Core Content:** The instructor presents open-ended questions, encouraging students to write their own campus epidemic prevention detection inventions in live chat or post-class forms. When designing, students are guided to prioritize ease of operation and applicability for all age groups, conveying scientific and humanistic care.
  - **Key Message:** Science is not limited to laboratory instruments; your creative ideas can also become epidemic prevention tools to protect those around you.

VI、Personnel Allocation and Division of Labor

- **General Supervisor (1 person):** Oversee course development, liaison with school authorities, coordination of online platforms, and overall risk management.
- **Online Lecturer (1 person):** Prepare script for explanations, record science popularization videos, and provide explanations and answer questions during the course.
- **Design Specialist (1 person):** responsible for creating tiered PowerPoint slides, popular science animations, and activity promotion posters.
- **Online Technical Assistants (2 persons):** Debug online platforms, collect questionnaires and creative forms.
- **Data Recording Specialist (1 person):** Compile statistics on video views and completion rates, organize questionnaires and student creative ideas, and write data reports.

VII、Materials List

- Promotional and Material Category:
  - Course Viewing Guide Document
- Equipment Category:
  - Laptop Computer, Projector
  - Camera/High-definition Smartphone

- Teaching Category:
- Hand-tiered popular science PowerPoint animation course
- Complete Script for Online Classroom Explanation
- Science Popularization Recorded Video
- Post-Class Online Questionnaire Template
- Creative Idea Collection Form
- Course Supporting Thinking Questions

Ⅷ、Risk Contingency Plan

- Platform Risks: Online video platform lagging, playback failure, link invalidation; simultaneously distribute video files to all classes and store videos via backup cloud drives.
- Content risk: The difficulty of the knowledge points is relatively high, making it hard for lower-grade students to comprehend. A simplified version of the PowerPoint and written popular science graphic content are provided as supplementary materials, feedback is collected, and content is adjusted accordingly.

Ⅸ、 Effect Evaluation

- Satisfaction Assessment: Collection of post-class online questionnaires and oral feedback from class tutors.
- Outcome Assessment: Count total video playbacks and valid viewers, aggregate student creative ideas, and produce a course summary report.